NewBear Computing Store Ltd

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77-68 Seventh Newsletter December 1979

Comment

In this issue we have a mixture of contributions. The most important issue, to my mind, is a letter which brings up the question of the standardisation of the 77-68 bus. Currently it is a fair bet that as each new board comes out its edge connector assignments will conflict with one of the other boards.

I was very pleased to receive instructions for making a MON 1 board have the same memory map as MON2- many thanks.

Mr. H.S. Lynes suggests a register of users equipment and interests is kept up to date, possibly on a regional basis. This seems a good idea but it takes effort. If someone would like to volunteer I will pass on any details I receive and we could concider how to publish the results when we see what the response is. On similar lines I am wondering whether we should run a library of 77-68 software. I have received a number of programs and articles but unfortunatly it is not possible to publish them because of space or because they are of marginal interest.

In view of my success at getting the MON 1 mods can I put forward another request? Has anyone worked out how to use the graphics capability of the VDU board from BASIC? Or tips for making PET BASIC programs work?

I am still receiving programs written in HEX. These are useless as they are difficult to understand and I have not the time to sort them out. Anything for publication must be either in symbolic form (or in S format if you want to keep your code secret).

O dear! The page numbers for the corrections to the dynamic RAM board were wrong. I was working from a Xerox and made unwarranted assumptions. For page 3 read page 7 and for page 7 read page 3. In addition on page 7 line 23 replace 'x18' by 'x18, x3' (correction to correction). On page 2 R9 and R10 are 2K7. On page 4 the capacitors in the 5v reulator should be C3 and C5 and are decoupling types.

The Floppy disc board and the 6809 boards will be available early in the new year, in the case of the disc very early.

Paul Bryant
The Bumbles

Well Meadow
Shaw
Newbury

List Of Suggestions

Below is a summary of suggestions I have received from Mr. J.L.Messenger with which I have conciderable sympathy and I publish them for comment from the members.

"There is a definite need for a code of practice to standardise the 77-68 bus. As well as defining edge connector assignments it should include suggestions on implementing a backplane". The need for this is to enable boards to be swapped around freely in a machine and also between users without the need to do surgery on the board or backplane. The current situations is a disaster as with each new board we have more problems to contend with. My view is that we should have standard edge assignments and all other connections should come out on the opposite edge on a second connector or for connection to a cable. I put forward my suggestion for comment.

1-2	0v
3	+5v
4	R/W
5	SEL
6	E
7	KEY WAY
8	HOLD
9	5 MHz
10	CLK
11	IRQ
12	NMI
13-15	SPARE
16-20	BANK ADDRESS
21-36	ADDRESS
37	0v
38-45	DATA
46-53	DATA SWITCHES
54-57	SPARE
58-66	BIT RATES
54-57	SPARE
72	-5v
73	RESET
74	+12v
75	-12v
76	+5v
77-78	0v

Note the inclusion of -5v as this is required on several boards and could be provided centrally. The inclusion of the bit rates will obviate the need to have these signals produced on more than one board. The 8 data switches are useful for injecting levels into either the CPU board or other boards.

Next Mr. Messenger wants 77-68 to expand. He wants more support from NewBear in the shape of a girl or two and a full time engineer, more boards but, perhapse, a higher subscription. Again I am sympathetic but one must remember that 77-68 is only part, a small part, of NewBears trade. Also 77-68 adicts tend to be economical and there is no fortune to be made. The investment that can be made must be related to the sales which are small compared to, say, NASCOM. In fact I have received no requests for new boards apart from the floppy disc one which is, in any case, comming soon. Suggestions are always welcome. Of cource the success of 77-68, to some extent, depends on the users who publish articles in the various magazines and show their wears at local ACC meetings and so forth.

John would like the newsletter on separate sheets of A4 for ease of filing. In fact the newsletter starts life on A4 and gets reduced for cheapness. Would you pay more for A4? Let me know and I'll see what can be done.

Lastly he suggests that perhapse the boards are not keeping up with technology. In particular he thinks a memory board based on MK4816 chips would be far superior to the 32K dynamic RAM board particularly in power consumption. Unfortunatly it takes many months to get a board designed, debugged, tracked and documented and thus one can not take advantage of new developments quickly. With the particular chip, MK4816, it does not seem to be easily available yet since I could not find an advert for it in the popular comics, UK or American.

I have been meditating.that perhapse we should run a user group meeting to discuss the future of 77-68, to swap ideas and perhapse put the group on a more democratic level by electing a committee. How much support have I and any suggestions for where and when?

Right, let's have your views.

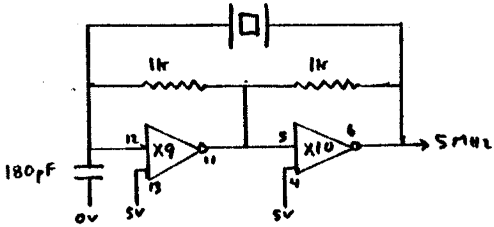
Do You Want Your 77-68 To Run At Double (& More) Speed?

Several constructors are running their machines with faster clocks. Typical is Roy Tipping.

The simple answere is to fit an appropriate crystal of up to 12 MHz insted of the 5 MHz supplied by NewBear. Don't be horror struck. I was trying to find a supplier of the MOSTEK 68B00, a 2MHz version of the ostensibly 1MHz 6800 when I heard that my colleagues at the BBC Research Department regularly use off-the-shelf 1MHz versions at up to 1.5MHz without any problems. In disbelief I bought a 10.7MHz crystal, and it works. All my programes run at 1.33MHz without any faults. The only aditional modification needed on my system, which uses all Bear Bags, was to put 4K7 pull up resistors on each of the 8 data lines on the backplane, and to make sure that all memory accesses were either to existing chips or only to the backplane.

To explain: My system has CPU, VDU, MON2 with T-BUG, 8K PROM BASIC (at C000-DFFF), Cottis Blandford CUTS interface (used mainly at 2400 bits/sec), 8x4K RAM boards, STAR device keyboard and 9" monitor. The C.S.S. BASIC clears all RAM and tests for 00 during initialisation. When running at 1MHz and above the buffered data lines, when a non existant memory location is accessed, drift at a voltage somewhere around 2v. The C.S.S. reads this as a succesful 00 and moves onto the next location. When it fails (@8000- the ACIA/PIA I/O area) it assumes that the whole of the bottom 32K is RAM and puts the Stack pointer at 7FFF. If you have 32K RAM so far so good, but anything less causes chaos. The pullups ensure that all empty addresses return FF. C.S.S. BASIC also tests @8004 for MIKBUGs PIA. Again a drifting bus may produce 00 which is the response of a PIA test. Rather than FFs which C.S.S. expects from empty sockets. Either use another 8x4K7 pull-ups or disable the chip enable decode on MON2.

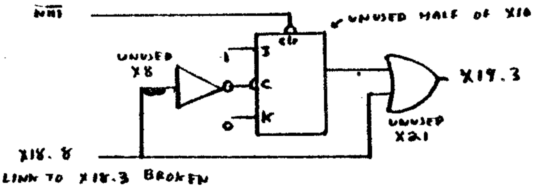
If you use a MON1 board then this has to be modified as the bit rates are determined from the 5MHz line. It's a mess trying to use the dividers to re-establish the correct rate, but it is possible to build an on-board crystal oscillator with 5MHz crystal you no longer need. As it happens there are two gates free on the board, on different chips, but adjacent to each other, so it is possible to build the oscillator on the board (though I built the discrete part of it off board on Veroboard). The circuit is below. Do not forget to disconnect the 5MHz to the board.



Roy Tipping

Unwanted Read From The Keyboard

The following circuit for the VDU board requires no new componants and gets rid of unwanted keyboard reads.



If no interrupt has been generated, then a read from any location which addresses the keyboard buffer will read from previously unused RAM locations. If NMI is low, then a keyboard read will access the 8212 (X24) removing the interrupt and setting the J-K Q output to logic 1.

If 2 devices can cause an NMI then the clear input to the J-K should be driven from the keyboard strobe via an inverter to ensure proper operation. The RAM location FBFF could then be used to determine which device had generated the NMI by storing in it a character which cannot be generated by the keyboard.

D.A. Roberts

Conversion Of MON1 To have Same Memory Map As MON2

This solution is not perfect, but it does work, and involves no extra hardware. When complete the memory map is

FC00-FFFF	1K	memory on MON 1 board
F800-FBFF	1K	memory map VDU
F400-F7FF	1K	echo of FC00-FFFF
E400-F3FF	4K	spare
E000-E3FF	1K	TBUG on PROM board
C000-DFFF	8K	spare
A000-BFFF	8K	256 byte CPU board memory plus echoes
8020-9FFF		echo of 8000-8020
8008-801F		free I/O addresses
8004-8007		echo of 8000-8003
8002-8003		ACIA B

8000-8001		ACIA A
2000-7FFF	24K	spare for memory.
0000-1FFF	8K	echos of 256 byte CPU memory (optional).

As can be seen, address decoding is far from perfect, but there is still 32K of contiguous memory from 0000-7FFF which is the best that any MIKBUK type system can have.

I have removed the 8212 interface from the VDU board and am at present introducing it to the PROM board so that it appears at 8010,8011.

The breaks and links required to achieve the above memory map are as follows:-

Cut tracks from	to	
x14 pin 8	x13 pin 15	
x12 pin 4	x12 pin 5	
x13 pin 1	x13 pin 9	
edge 31	x13 pin 13	
edge 32	x13 pin 14	
'c'	x13 pin 12	*

Remove link from	to	
x13 pin 9	x12 pin 1	
x11 pin 13	x4 pin 12	
x5 pin 11	x14 pin 14	
'a'	'c'	*

Add link from	to
x14 pin 13	x13 pin 13
x14 pin 12	x13 pin 15
x14 pin 10	x13 pin 14
x14 pin 8	x13 pin 1
x13 pin 1	x12 pin 1
x11 pin 13	x4 pin 10
x29 pin 11	x5 pin 11
edge 31	x12 pin 4
x13 pin 9	'a'

* Indicates that this alteration is only required if the 256 byte CPU board RAM is not to lie at 0000-1FFF

Simon Sweeney

Progress Report From Germany

Just before the summer holiday, the "Local User Group" in Hamburg had two completed 4K 77-68 systems running. These were made from

Bear-Bags or just PCBs from various UK sources ("next time you're in town, will you get Bear-Bag N from....") The German speaking users are working from a translation of the design manual recently completed by Peter Bendall and Bernd Robrahn. To simplify building without VERO parts, Bernd designed Front Panel and Mother Board PCBs. Both existing systems have BUG-1 or BUG-2 loaded into MON-1 via the Cottis-Blandford Cassette interface and a standard cassette has been made with both monitors and binary dump and load in the top 1K of memory. Because the cassette interface is TTL, unlike the 'standard' interface which is V24, the BUG-1 uses ACIA'b' as a terminal. Bernd is using a 20mA Petitevid that belonged to Peter as well as improving BUG-2.

Both systems have the 4 'basic' boards, CPU MON-1, 4K RAM and VDU. Both VDU's exhibit identical instability which was cured by patching a 2nF capacitor across pin 11 of X-11 and ground.

A copy of Tom Pittman's TINY BASIC ran as soon as the patches JMP GEPCH and PNTCH were loaded in locations 106 and 109, but only under BUG-1. In order to run BUG-2 and the VDU, the control characters, fillers and so on, have to be patched out (Tom gives particulars in the manual). Once patched it will run either system BUG.

A simple Editor has been written which allows up to 1K of text to be edited in the memory, and then stored on tape. It is hoped that, by the end of the summer, it will be possible to assemble the contents of the store buffer, writing the binary to tape.

We discovered (not entirly by accident) that coupling a V24 or 20mA interface Kansas City Interface and a tape recorder to any 300 bit/sec terminal socket, like a PDP11 or even a Time-Sharing system, without the clock connected, still works. That means that we can use a cross assembler on a larger system which has been a great help.

Peter's EPROM board (ROM-A) is complete, as is the 32K dynaram, but the 4116's ordered from Priority One still have not arrived. Until then BASIC is just Tiny. A set of 6 BASIC games were written for TINY BASIC and they can be loaded and saved by USR(0) and USR(3) functions which use the contents of 0020/1 and 0024/5 as the first and last addresses to save.

The EPROM programer from Doctor Dobbs (circuit in the latest 6800 Micro-Journal) is ready to try, just as soon as we get the software written. similarly we have proved that, with suitable choice of clock, we can produce 5 unit code for RTTY output by setting the three upper bits to 1. At 50 bits/sec it runs about 2/3 normal speed. You can also read it in but only by carefully typing. That means that a version of BUG-1 would be easy. The system wouldn't work to translate RTTY off the air though because it needs at least 4 1/2 stop bits. TASS wouln't want to change it for you.

Contact address for German Speaking 77-68 users, and for others interested is:-

Peter Bendall
Flottmoorring 67
2358 Kaltenkirchen

Store Tests

The store test in the 77-68 manual is defficient in that some types of board and chip faults can not be detected. The problems are:-
1. The repetition period is a power of 2
2. The addresses are on the bus for a long time with indexing.
3. The repetitive patern, particularly on the least significant bits, will not detect some chip faults.

The program below overcomes these problems and also is a good CPU tester. This was of interest to me as I used to run my CPU at 13.33MHz and I found one memory chip started to fail at that speed. With the dynaram the speed had to be cut to 10.25 MHz- so it no longer runs rings round a 4MHz Z80 but still good.

```
                          NAM    STES
                   *STORE TEST PROGRAM
FF80                      ORG    $FF80
FF80 0002          SLO    RMB    2
FF82 0002          SHI    RMB    2
FF84 0001          RG1    RMB    1
FF85 0001          RG2    RMB    1
FF86 0001          INST   RMB    1
FF87 0002          INAD   RMB    2
FF89 0001          INRE   RMB    1
FF8A B6 FF84 GNV   LDA A  RG1        PSEUDO RANDOM
FF8D 84 1D         AND A  #$1D       NUMBER GENERATOR
FF8F 5F            CLR B
FF90 46            ROR A
FF91 C9 00         ADC B  #0
FF93 46            ROR A
FF94 46            ROR A
FF95 C9 00         ADC B  #0
FF97 46            ROR A
FF98 C9 00         ADC B  #0
FF9A 46            ROR A
```

```
00022 FF9B C9 00            ADC B  #0
00023 FF9D 56               ROR B
00024 FF9E B6 FF84          LDA A  RG1
00025 FFA1 46               ROR A
00026 FFA2 B7 FF84          STA A  RG1
00027 FFA5 39               RTS
00028 FFA6 FF FF87 SV       STX    INAD      MODIFY EXTENDED INSTRUCT
00029 FFA9 7E FF86          JMP    INST
00030                *ENTRY POINT
00031 FFAC 7F FF85 MTST2    CLR    RG2
00032 FFAF B6 FF85 MTCY     LDA A  RG2
00033 FFB2 4A               DEC A
00034 FFB3 B7 FF84          STA A  RG1
00035 FFB6 B7 FF85          STA A  RG2
00036 FFB9 FE FF80          LDX    SLO       LOWER TEST LIMIT
00037 FFBC BC FF82          CPX    SHI       UPPER TEST LIMIT+1
00038 FFBF 26 02            BNE    MTWL
00039 FFC1 3F               SWI
00040 FFC2 FE               FCB    $FE
00041 FFC3 86 B7   MTWL     LDA A  #$B7      WRITE LOOP
00042 FFC5 B7 FF86          STA A  INST
00043 FFC8 86 39            LDA A  #$39
00044 FFCA B7 FF89          STA A  INRE
00045 FFCD 8D BB   MTWL1    BSR    GNV       SET NEXT
00046 FFCF 8D D5            BSR    SV        STORE IT
00047 FFD1 08               INX
00048 FFD2 BC FF82          CPX    SHI       AND LOOP
00049 FFD5 26 F6            BNE    MTWL1
00050 FFD7 FE FF80          LDX    SLO
00051 FFDA 86 F6            LDA A  #$F6      SET UP FOR READING
00052 FFDC B7 FF86          STA A  INST
00053 FFDF B6 FF85          LDA A  RG2
00054 FFE2 B7 FF84          STA A  RG1       READ LOOP
00055 FFE5 8D A3   MCHK     BSR    GNV       GET NEXT
00056 FFE7 8D BD            BSR    SV        AND RETRIEVE STORED
00057 FFE9 11               CBA
00058 FFEA 26 0B            BNE    MT2E      IF NE THEN ERROR
00059 FFEC 08               INX              ELSE LOOP
00060 FFED BC FF82          CPX    SHI
00061 FFF0 26 F3            BNE    MCHK
00062 FFF2 7C FBC7          INC    $FBC7     INDICATE PASS COMP
00063 FFF5 20 B8            BRA    MTCY      RECYCLE WITH NEW SEED
00064 FFF7 3F      MT2E     SWI
00065 FFF8 FE               FCB    $FE
00066                       END

SYMBOL TABLE

SLO    FF80  SHI    FF82  RG1    FF84  RG2    FF85  INST   FF86
```

INAD	FF87	INRE	FF89	GNV	FF8A	SV	FFA6	MTST2	FFAC
MTCY	FFAF	MTWL	FFC3	MTWL1	FFCD	MCHK	FFE5	MT2E	FFF7

Andy Holt

Advert

For Sale.

77-68 CPU board. All chips in sockets and working #65. 4K Ram. All chips in sockets and working #80. VDU board. Chips in sockets. Board has a bug #70. MON-2 board fitted Swatbug and 2 PIA's. Has 5v s/c on board. #85.

Archie Douglas
48 Marmion Drive
Glenrothes
Fife
KY62PF
Tele (0592) 753149

An Offer For a Deserving Case

Congratulations on a fine newsletter. (ED. Thanks!) However, I am now too old. (Hang gliding maybe but computing? Never!). If you have a deserving case such as an impecunious youngster perhapse, I will gladly send you my 77-68 board.

Anon

ED. A generous offer. To save Anon from making a profit from the sale of wast paper please send me your nominations and Anon and I will arbitrate.

Graphics and a Hard Monitor for the MON 1

Simple graphics can be implemented with 2 ICs. This is done by adding a second character gererator to the board in parallel with the 74S262 ROM. It uses a 1K x 8 EPROM of the 2758 type (or 27L08) and a 74LS03 open collector NAND chip. These are located on the PCB on the prototyping space and together with some unused gates are wired with links on the reverse of the board.

The circuit has two modes of operation, set by a sub-min. toggle switch glued to the front of the board. In the lower position the orriginal operation of the circuit is unaffected. In the upper position, the inverted video characters are replaced by the graphics symbols while the non-inverted alpha numeric characters are unaffected. This makes it possible to mix graphics and alpha numerics on the same display.

The graphics are semigraphics and comprise the 64 TELETEXT characters plus 32 miscellaneous and their inverses. In my PROM, these are chess pieces, the four suits, graph plotting bits, maze drawing bits and a few miscellaneous symbols. All characters are contiguous.

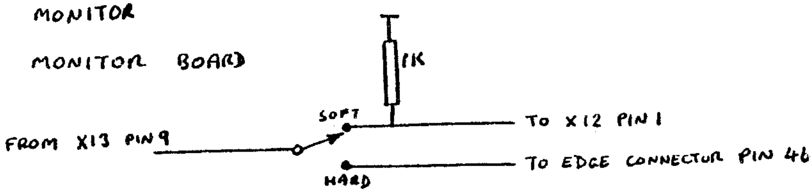
The program supplied may be used to generate others for custom ROMs. The diagram shows the circuit diagram of the unit.

A hard monitor may be added to MON 1. It uses a 2708, a 74LS244 (or 81LS97 etc) and a 74LS00. The chips are assembled on a prototyping board (I used the one with my CUTS CCT) and is enabled from the monitor board by pin 46 on the backplane. Another switch on the front of the board selects hard or soft monitor. When in hard mode, the existing RAM may still be written to if the write protect is removed. This makes the hard monitor very easy to implement.

Dave Roberts

HARD MONITOR

(1) TO MONITOR BOARD



(2) ON PROTO. BOARD

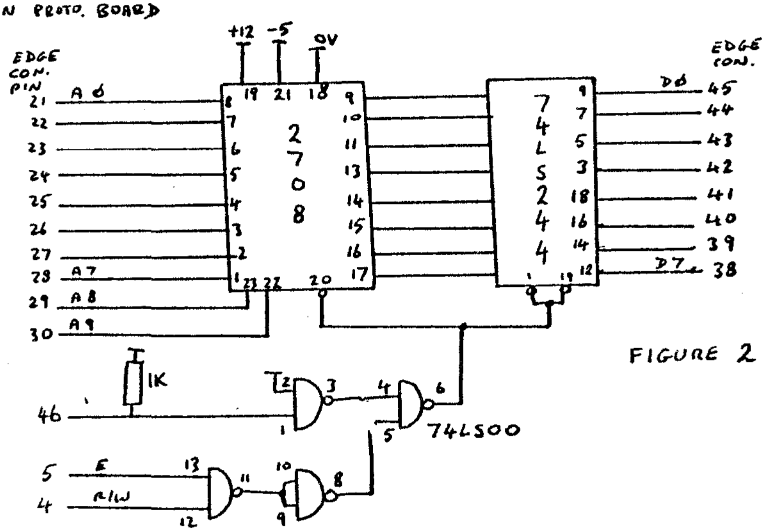
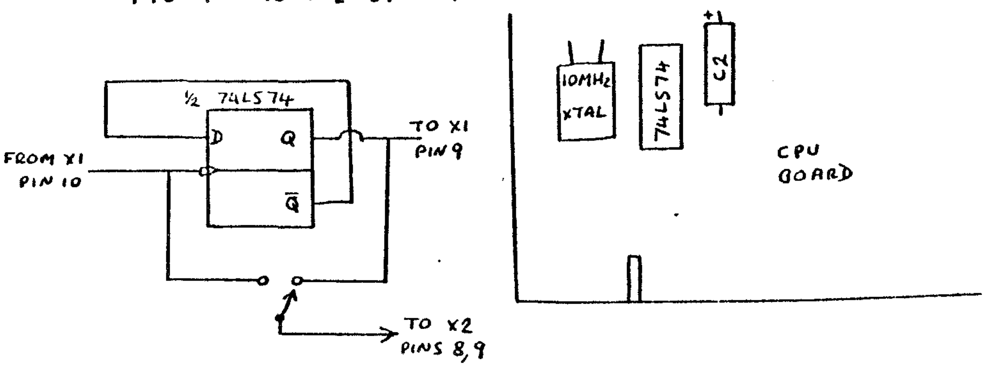
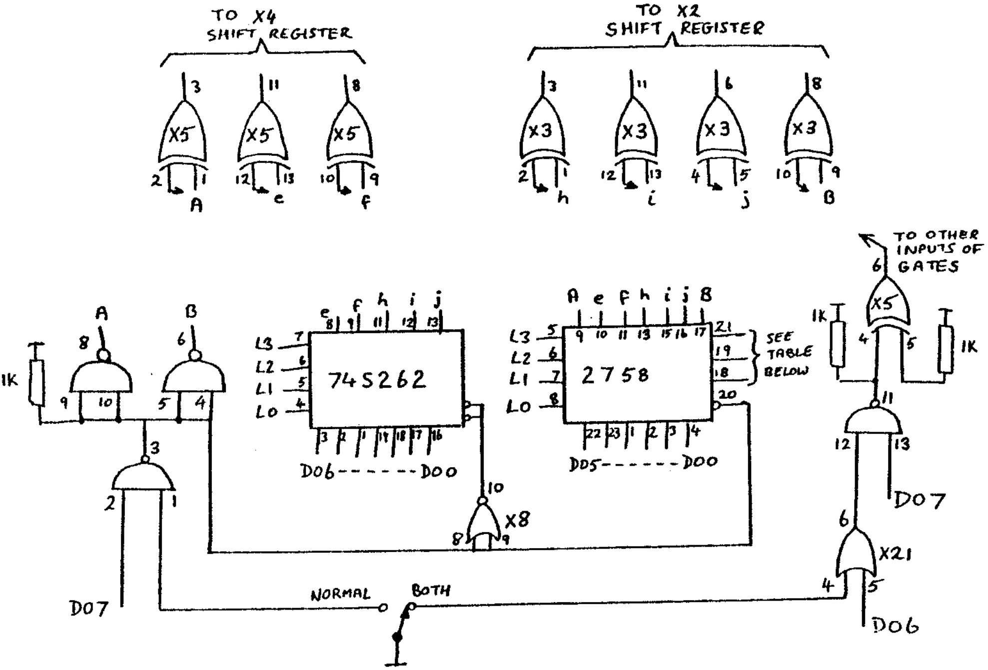


FIGURE 2

FIG 4 - 10MHz SYSTEM



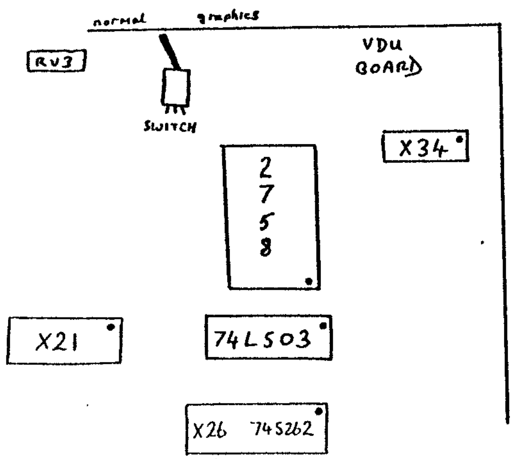
GRAPHICS CIRCUIT



UNMARKED GATES ARE THE 74LS03

PROM PINS

		18	19	21
PROM TYPE	2758	PIN 20	GND	+5V
	2708	GND	+12V	-5V



NOTE

POWER DISSIPATION MEANS THAT THE ROM MUST BE A 2758 OR 27L08 RATHER THAN 2708. THE NECESSARY +12V AND -12V ARE AVAILABLE ON UNUSED PINS ON THE EDGE CONNECTOR (MON-1 BUS STRUCTURE). A -5V REGULATOR (79L05) MUST BE PROVIDED FOR THE 27L08

```
MON < G 2000 START 1000 END 13FF
S11310000101010101010101010FFFFFFFFFFF42
S113101 0F0F010101010101010F0FFFFFFFFFFFFF92
S11310201F1F101010101010101FFFFFFFFFFFFFF5
S1131030FFFF10101010101010FFFFFFFFFFFFF45
S11310401010F0F0F01010101010FFFFFFFFFFFFF62
S1131050F0F0F0F0F01010101 0F0FFFFFFFFFFFFFB2
S113106 01F1FF0F0F01010101 01FFFFFFFFFFFFF15
S1131070FFFFF0F0F010101010FFFFFFFFFFFFFF65
S113108010101F1F1F101010101 0FFFFFFFFFFFFF95
S1131090F0F01F1F1F1010101 0F0FFFFFFFFFFFFFE5
S11310A01F1F1F1F1F10101010 1FFFFFFFFFFFFFF48
S11310B0FFFFF1F1F1F10101010FFFFFFFFFFFFFFF98
S11310C0101 0FFFFFF10101010 10FFFFFFFFFFFFFB5
S11310D0F0F0FFFFFF1010101 0F0FFFFFFFFFFFFF05
S11310E01F1FFFFFFF101010101FFFFFFFFFFFFFF68
S11310F0FFFFFFFFFFF10101010FFFFFFFFFFFFFFB8
S1131100010101010 0F0F0F0F010FFFFFFFFFFFFFC1
S11311 10F0F010101 0F0F0F0F0F0FFFFFFFFFFFFF11
S11311201F1F101010 0F0F0F0F01FFFFFFFFFFFFF74
S1131130FFFF101010F0F0F0F0FFFFFFFFFFFFFFC4
S1131140101 0F0F0F0F0F0F0F010FFFFFFFFFFFFE1
S1131150F0F0F0F0F0F0F0F0F0F0FFFFFFFFFFFFF31
S11311601F1FF0F0F0F0F0F0F01FFFFFFFFFFFFF94
S1131170FFFFF0F0F0F0F0F0FFFFFFFFFFFFFFFE4
S11311801010 1F1F1FF0F0F0F010FFFFFFFFFFFFF14
S1131190F0F01F1F1FF0F0F0F0F0FFFFFFFFFFFFF64
S11311A01F1F1F1F1FF0F0F0F01FFFFFFFFFFFFFC7
S11311B0FFFFF1F1F1FF0F0F0F0FFFFFFFFFFFFFFF17
S11311C01010FFFFFFF0F0F0F010FFFFFFFFFFFFF34
S11311D0F0F0FFFFFFFF0F0F0F0F0FFFFFFFFFFFFF84
S11311E01F1FFFFFFFFF0F0F0F01FFFFFFFFFFFFFE7
S11311F0FFFFFFFFFFFF0F0F0F0FFFFFFFFFFFFFFF37
S11312003810101010101010 1038FFFFFFFFFFFFF0
S1131210383810101010101010 10FFFFFFFFFFFFFE0
S1131220103838101010101010 10FFFFFFFFFFFFD0
S113123010103838101010101010FFFFFFFFFFFFC0
S113124010101038381010101010FFFFFFFFFFFFB0
S113125010101010383810101010FFFFFFFFFFFFA0
S1131260101010101038381010 10FFFFFFFFFFFF90
S113127010101010101038381010FFFFFFFFFFFF80
S11312801010101010101038381 0FFFFFFFFFFFF70
S11312901010101010101010FFFF10FFFFFFFFFFFFD2
S11312A0D0D0D0D0D0D0D0D0D0D0D0FFFFFFFFFFFF20
S11312B0FF10101010101010101 0FFFFFFFFFFFFFB2
S11312C01313131313131313131 3FFFFFFFFFFFF62
S11312D0D0D0D0D0D0D0D0D0FFFFD0FFFFFFFFFFFF92
S11312E0FFD0D0D0D0D0D0D0D0D0FFFFFFFFFFFFFF82
S11312F0FF131313131313131 3FFFFFFFFFFFFFF5A
S1131300131313131313 13FFFF13FFFFFFFFFFFF49
S113131076FFFFFF7E3C18101010FFFFFFFFFFFF5A
S1131320 3C3CDBFFDB3C7E101010FFFFFFFFFFFFA8
S1131330183C7EFF7E3C1810101 0FFFFFFFFFFFDC
S1131340181 83C7EFFFF7E181010FFFFFFFFFFFF01
S11313503870383838387C7C1010FFFFFFFFFFFFE3
S11313605A7E3C3C3C7E7E1010FFFFFFFFFFFFF9B
S1131370387CFE5E1E3C7F7E1010FFFFFFFFFFFFE9
S11313801818 3C3C3C187E7E1010FFFFFFFFFFFF47
S1131390185A5A3C3C3C7E7E1010FFFFFFFFFFFFD3
S11313A03C7E7E3C3C3C7E7E1010FFFFFFFFFFFF37
S11313B03C3C3CFFFF3C3C3C3C3CFFFFFFFFFFFFF51
S11313C0131614181830 70D09011FFFFFFFFFFFFA1
S11313D0D0F0F0F8F8FCFCFEFF90FFFFFFFFFFFFEA
S11313E013171 71F1F3F3F7FFF11FFFFFFFFFFFF73
S11313F0D07030181814161311 90FFFFFFFFFFFF71
MON < E PORT-2C
```

CONTIGUOUS TELETEXT (INVERTED VERSIONS FORM THE OTHER 32)

GRAPH PLOTTING

MAZE PLOTTING

HEARTS
CLUBS
DIAMONDS
SPADES

SUITS

PAWN
ROOK
KNIGHT
BISHOP
QUEEN
KING

CHESS

+
/

TRY THIS

```
10 HOME
20 FOR I = 0 TO 1020
30 POKE(63488+I, RND
   +169)
40 NEXT I
50 WAIT 255
```

```
                        NAM  GRAPHICS DESIGN
                *
                * THIS PROGRAM IS USED TO DESIGN
                * GRAPHICS CHARACTERS FOR VDU 1
                * BOARD. ON ENTRY , THE USER IS
                * PROMPTED FOR A CHARACTER LOCN
                * THIS IS A HEX LOCN IN THE RANGE
                * 00 - 3F  AND CORRESPONDS TO THE
                * 64 POSSIBLE CHARACTER NUMBERS IN
                * THE ROM. THE CURSOR WILL THEN BE
                * POSITIONED IN THE LEFT HAND COL.
                * AND THE DISPLAY MAY BE FORMED.
                * IF A SPACE IS ENTERED, A '0' WILL
                * BE PROGRAMMED INTO THAT BIT POSITION
                * IF A "DEL" IS ENTERED, A '1' WILL
                * BE PROGRAMMED.
                * AFTER A ROW OF SEVEN HAS BEEN
                * ENTERED,CORRESPONDING TO THE SEVEN
                * COLUMNS IN THE DISPLAY,CR-LF WILL
                * BE SENT FOR THE NEXT ROW.
                * TEN ROWS ARE ENTERED, AND THEN THE
                * PROGRAM RETURNS TO THE START.
                * THE DATA PRODUCED IN THE RANGE
                * $1000-$13FF MAY THEN BE PROGRAMMED
                * INTO ROM.IF YOU USE A NON INTERRUPT
                * DRIVEN MONITOR FOR INEEE THEN
                * YOU'LL HAVE TO ALTER THE PROGRAM.
                * THE REASON FOR THE EXTRA BIT IN
                * THE MIDDLE OF THE BYTE INSTEAD OF
                * AT THE END IS DUE TO THE USE OF
                * A ROM IN THE ORIGINAL WHICH HAD
                * ONE OF ITS OUTPUT LEGS MISSING.
                * NOTE THAT THE OTHER 64 CHARACTERS
                * ARE FORMED AS THE INVERSES OF THE
                * 64 IN THE ROM. HENCE BLACK AND
                * WHITE CHESS PIECES ETC.
                * FILL THE MEMORY BLOCK WITH $FF
                * BEFORE STARTING , AND DO NOT
                * BACKSPACE PAST THE START OF A
                * LINE AS THIS WILL PRODUCE ERRORS
                *
                *
1800                    ORG  $1800
                        OPT  O,H,X
                *
1800 7E 1815    GRAPHICS JMP  START      ENTRY HERE
1803 7E FE55    IN2HS    JMP  BYTE       (2 HEX INPUT)
1806 7E FECC    SPACEOUT JMP  OUTS
1809 7E FEC8    OUT4HEX  JMP  OUT4HS
180C 7E FE7E    P.STRING JMP  PSTRING
180F 7E FFD1    OUTPUT   JMP  OUTEEE
1812 7E EDBB    CR.LF    JMP  CRLF
                *
                *
1815 8E 18F0    START    LDS  £$18F0
1818 CE 18A8             LDX  £PROMPT
```

```
181B 8D EF (180C)        BSR  P.STRING
181D 8D E4 (1803)        BSR  IN2HS      GET CHAR. NUM.
181F C6 10               LDAB £$10
1821 48                  ASLA            MULTIPLY
1822 48                  ASLA            NUMBER
1823 48                  ASLA            BY 8
1824 24 02 (1828)        BCC  :0         IN FIRST BLOCK
1826 CA 02               ORAB £2
1828 48           :0     ASLA
1829 24 02 (182D)        BCC  :1
182B CA 01               ORAB £1
182D B7 18C0      :1     STAA TEMP.VAR+1
1830 F7 18BF             STAB TEMP.VAR
1833 8D D1 (1806)        BSR  SPACEOUT
1835 CE 18BF             LDX  £TEMP.VAR
1838 8D CF (1809)        BSR  OUT4HEX    PRINT FULL ADDRESS
183A 8D D6 (1812)        BSR  CR.LF
                  * START OF LOOPING
                  * A CONTAINS LINE COUNT
                  * B CONTAINS COL. COUNT
183C 86 0A               LDAA £$A
183E C6 20        :11    LDAB £$20
1840 7F 18C1             CLR  FLAG
1843 36                  PSHA
1844 3E           :2     WAI             INPUT FROM KEYBOARD
1845 B6 F000             LDAA INBUF      KBD WRITES HERE
1848 81 7F               CMPA £$7F       IS IT 'DEL'
184A 27 1F (186B)        BEQ  :3
184C 81 20               CMPA £$20       IS IT A SPACE
184E 27 22 (1872)        BEQ  :4
1850 81 08               CMPA £8         BACKSPACE ?
1852 26 F0 (1844)        BNE  :2
1854 8D B9 (180F)        BSR  OUTPUT
1856 7D 18C1             TST  FLAG
1859 26 03 (185E)        BNE  :22
185B 54           :21    LSRB
185C 20 E6 (1844)        BRA  :2
185E C5 E0        :22    BITB £$E0
1860 26 F9 (185B)        BNE  :21
1862 54                  LSRB
1863 54                  LSRB
1864 CA 80               ORAB £$80
1866 7F 18C1             CLR  FLAG
1869 20 D9 (1844)        BRA  :2
186B 86 A0        :3     LDAA £$A0
186D 8D A0 (180F)        BSR  OUTPUT
186F 0D                  SEC             SET BIT TO '1'
1870 20 03 (1875)        BRA  :5
1872 8D 92 (1806) :4     BSR  SPACEOUT
1874 0C                  CLC             SET BIT TO '0'
1875 59           :5     ROLB            ADD NEW BIT TO LINE
1876 24 CC (1844)        BCC  :2         RECD. 3 YET
1878 7D 18C1             TST  FLAG       DONE A WHOLE LINE
187B 26 08 (1885)        BNE  :6         YES,THEN BRANCH
187D 7C 18C1             INC  FLAG
1880 59                  ROLB
```

```
0113                         * SET UP FOR ANOTHER 4 BITS
0114                         * AND SET MIDDLE BIT TO '1'
0115 1881 CA 11                        ORAB £$11
0116 1883 20 BF (1844)                 BRA  :2
0117 1885 FE 18BF            :6        LDX  TEMP.VAR
0118 1888 E7 01                        STAB 1,X        STORE BYTE
0119 188A 08                           INX             MOVE POINTER
0120 188B FF 18BF                      STX  TEMP.VAR
0121 188E 8D 82 (1812)                 BSR  CR.LF
0122 1890 32                           PULA
0123 1891 4A                           DECA
0124 1892 26 AA (183E)                 BNE  :11        INPUT 10 LINES
0125 1894 B6 18C0                      LDAA TEMP.VAR+1
0126 1897 80 0A                        SUBA £$A        MOVE POINTER BACK
0127 1899 B7 18C0                      STAA TEMP.VAR+1 TO FIRST LINE
0128 189C FE 18BF                      LDX  TEMP.VAR   DUE TO HARDWARE
0129 189F E7 00                        STAB X          WHICH PUTS LINE 10
0130 18A1 C6 FF                        LDAB £$FF
0131 18A3 E7 0A                        STAB $A,X
0132 18A5 7E 1815                      JMP  START      AT THE TOP
0133                         *
0134                         *
0135 18A8 0D0A               PROMPT    FCB  $D,$A
0136 18AA 4E4558542043                 FCC  'NEXT CHARACTER LOCN.'
0137 18BE 04                           FCB  4
0138                         *
0139                         *
0140 18BF                    TEMP.VAR  RMB  2
0141 18C1                    FLAG      RMB  1
0142                         *
0143                         *
0144                         * MONITOR ALLOCATIONS
0145                         *
0146 FE55                    BYTE      EQU  $FE55
0147 FECC                    OUTS      EQU  $FECC
0148 FFD1                    OUTEEE    EQU  $FFD1
0149 EDBB                    CRLF      EQU  $EDBB
0150 FE7E                    PSTRING   EQU  $FE7E
0151 FEC8                    OUT4HS    EQU  $FEC8
0152                         *
0153 F000                    INBUF     EQU  $F000


BYTE      FE55  0047 0146
CR.LF     1812  0052 0073 0121
CRLF      EDBB  0052 0149
FLAG      18C1  0079 0090 0099 0109 0111 0141
GRAPHICS  1800  0046
IN2HS     1803  0047 0058
INBUF     F000  0082 0153
OUT4HEX   1809  0049 0072
OUT4HS    FEC8  0049 0151
OUTEEE    FFD1  0051 0148
OUTPUT    180F  0051 0089 0102
OUTS      FECC  0048 0147
P.STRING  180C  0050 0057
PROMPT    18A8  0056 0135
PSTRING   FE7E  0050 0150
SPACEOUT  1806  0048 0070 0105
START     1815  0046 0055 0132
TEMP.VAR  18BF  0068 0069 0071 0117 0120 0125 0127 0128 0140
```